

Double-click (or enter) to edit

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
plt.style.use('ggplot')
```

```
#& importing file
df1= pd.read_csv('datasets.csv')
```

```
df= df1.copy()
```

```
df
```


	id	name	host_id	host_name	neighbourhood_group	neighbourhood	lat
0	1.312228e+06	Rental unit in Brooklyn · ★5.0 · 1 bedroom	7130382	Walter	Brooklyn	Clinton Hill	40.68
1	4.527754e+07	Rental unit in New York · ★4.67 · 2 bedrooms · ...	51501835	Jeniffer	Manhattan	Hell's Kitchen	40.74
2	9.710000e+17	Rental unit in New York · ★4.17 · 1 bedroom	528871354	Joshua	Manhattan	Chelsea	40.74

df.info()

2

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 20770 entries, 0 to 20769

Data columns (total 22 columns):

Column Non-Null Count Dtype

0 id 20770 non-null float64

1 name 20770 non-null object

2 3 host_id 20770 non-null int64

3 host_name 20770 non-null object

4 neighbourhood_group 20770 non-null object

5 neighbourhood 20763 non-null object

6 latitude 20763 non-null float64

7 longitude 20763 non-null float64

8 room_type 20763 non-null object

9 4 price 20763 non-null float64

10 minimum_nights 20763 non-null float64

11 number_of_reviews 20763 non-null float64

12 last_review 20763 non-null object

13 reviews_per_month 20763 non-null float64

14 calculated_host_listings_count 20763 non-null float64

15 availability_365 20763 non-null float64

16 number_of_reviews_ltm 20763 non-null float64

17 license 20770 non-null object

18 20765 rating 20770 non-null object

19 bedrooms 20770 non-null object

20 beds 20770 non-null int64

21 baths 20770 non-null object

20765

2.473690e+07

· ★4.75 · 1 bedroom

186680487

Henry D

Manhattan

Lower East Side

40.71

df.isna().any().sum()

np.int64(12)

df.sample(10)

20767

5.182527e+07

New York
· ★4.93 · 1 bedroom

304317395

Jeff

Manhattan

Hell's Kitchen

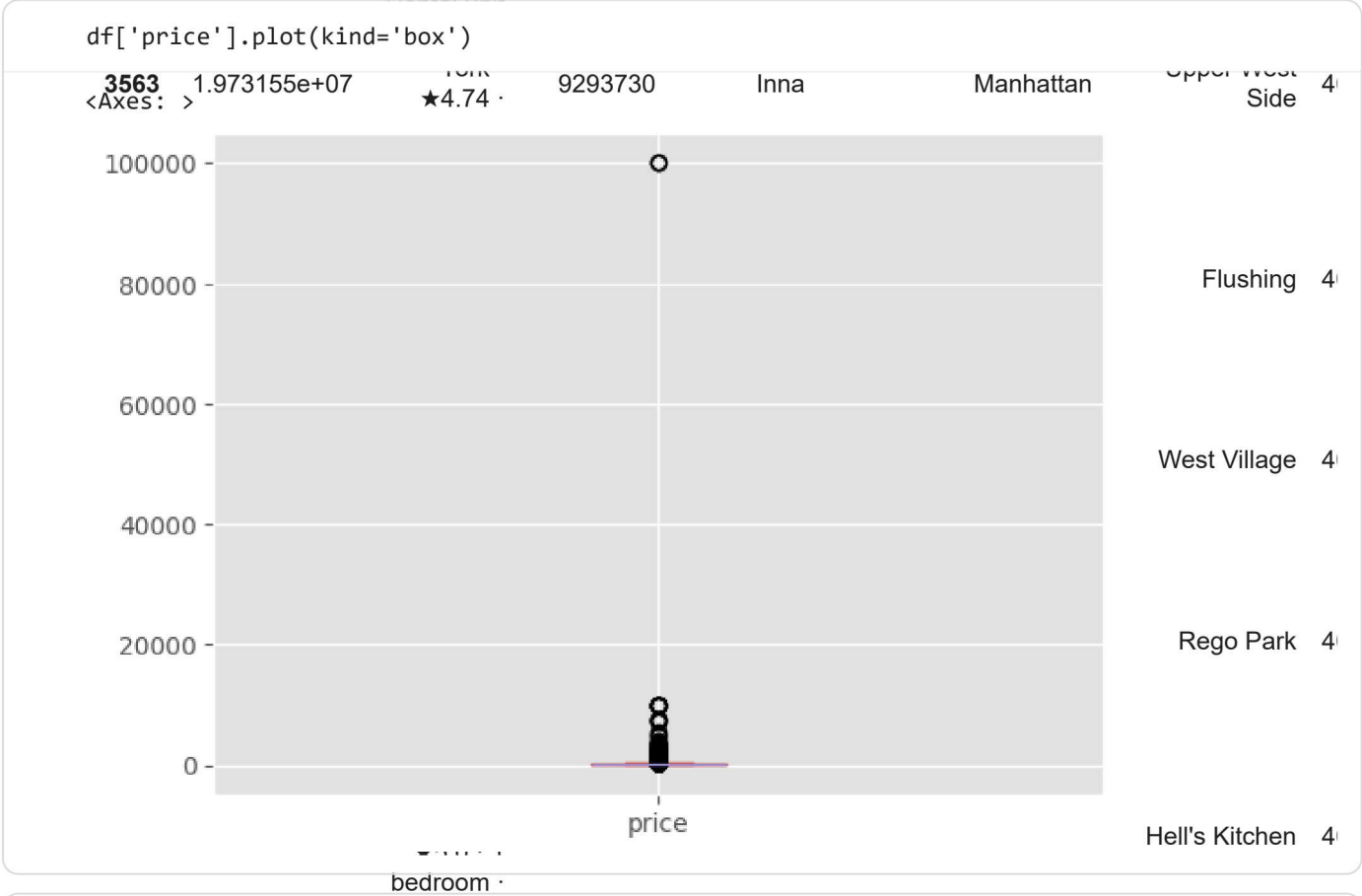
40.74

20768	7.830000e+17	Rental unit in New York · ★5.0 · 1 bedroom · 1...	163083101	Marissa	Manhattan	Chinatown	40.
20769	5.660000e+17	Rental unit in Queens · ★4.89 · 1 bedroom · 1 ...	93827372	Glenroy	Queens	Rosedale	40.

20770 rows × 22 columns

	id	name	host_id	host_name	neighbourhood_group	neighbourhood	1
	Home in						
df.shape							
(20770, 22)	bedroom · 1 bed · ...						

#! data describe df.describe()							
	2 beds ...						
	id	host_id	latitude	longitude	price	minimum_nights	r
	Home in						
count	2.077000e+04	2.077000e+04	20763.000000	20763.000000	20736.000000	20763.000000	
mean	3.033858e+17	1.749000e+08	40.726821	-73.939179	187.714940	20.000493	
std	3.901221e+17	1.725657e+08	0.060293	0.061403	1023.245124	33.532697	
min	2.595000e+03	1.678000e+03	40.500314	-74.249840	10.000000	1.000000	
25%	2.707260e+07	2.041184e+07	40.684159	-73.980755	80.000000	30.000000	
50%	4.992852e+07	1.086990e+08	40.722890	-73.949597	125.000000	30.000000	
75%	7.220000e+17	3.143997e+08	40.763106	-73.917475	199.000000	30.000000	
max	1.050000e+18	5.504035e+08	40.911147	-73.713650	100000.000000	1250.000000	



df[df['price']>1000]

10 rows × 22 columns

```
df[df['room_type']=='Private room']['price']
```

```
0      55.0
3     120.0
8     120.0
13     90.0
15     120.0
...
20758    65.0
20759    53.0
20762    79.0
20765    45.0
20769   102.0
Name: price, Length: 8803, dtype: float64
```

```
df['price'].skew()
```

```
np.float64(89.97557856732496)
```

```
## data cleaning
df.isnull().sum()
```

```
id                0
name              0
host_id           0
host_name         0
neighbourhood_group  0
neighbourhood      7
latitude           7
longitude          7
room_type         7
price            34
minimum_nights     7
number_of_reviews  7
last_review        7
reviews_per_month  7
calculated_host_listings_count  7
availability_365   7
number_of_reviews_ltm  7
license           0
rating            0
bedrooms          0
beds              0
baths             0
dtype: int64
```

```
df[df['latitude'].isnull()]
```

	id	name	host_id	host_name	neighbourhood_group	neighbourhood	lat
	15913	Guest suite in New York · ★4.88 · 8 bedrooms · ...	380290574	Karen	Manhattan	NaN	
	482	Loft in New York · ★4.48 · 1 bedroom · ...	46941561	Maria	Manhattan	Upper East Side	4
	15914	Brooklyn · 1... bedroom Rental unit in New York · 4	51954926	K	Brooklyn	NaN	
	579	Rental unit in New York · 4	108712614	Caroline	Manhattan	Upper West	4
	15915	Rental unit in New York · ★4.90 · 1 bedroom · ...	2321870	David	Manhattan	NaN	
	19156	Loft in Brooklyn · ★4.89 · 1 bedroom · 1 bed · ...	12385603	Yuichi Anne	Manhattan Brooklyn	Hell's Kitchen	4
...	...	...	...	...	...	...	...
	15917	Rental unit in Brooklyn · ★4.81 · 1 bedroom · ...	16392320	Oscar	Brooklyn	NaN	
	19159	Brooklyn · ★4.77 · 14 bedrooms · ...	10416706	Tzvi	Brooklyn	Sea Gate	4
	15918	Brooklyn · ★4.80 · 2 Townhouse bedrooms in New York ·	67118749	Afriqiyah	Brooklyn	NaN	
	19160	7 3000000e+17 · 11176320 · Jan			Manhattan	Upper East	4
	15919	Rental unit in New York · ★4.50 · Studio · 2 b... bedrooms · 5 beds · ...	348619646	Nomad	Manhattan	NaN	
7 rows × 22 columns							
	20687	Boutique hotel in New York · ★5.0 · 1	26556695	Justin And Alyssa	Manhattan	Midtown	4

drop the null values

167 rows × 22 columns

```
df.dropna(subset=['latitude','last_review','minimum_nights','longitude','price'],inplace=True)
```

```
df.isnull().sum()
```

```
id          0
name        0
host_id     0
host_name   0
neighbourhood_group  0
neighbourhood  0
latitude    0
longitude   0
room_type   0
price       0
minimum_nights  0
number_of_reviews  0
last_review  0
reviews_per_month  0
calculated_host_listings_count  0
availability_365  0
number_of_reviews_ltm  0
license     0
rating      0
bedrooms    0
beds        0
baths       0
dtype: int64
```

```
df.shape
```

```
(20736, 22)
```

```
#! dealing with duplicated records
df[df.duplicated()]
```

```
#! delete thee duplicate value
df.drop_duplicates(inplace=True)
```

```
df.duplicated().sum()
```

```
np.int64(0)
```

```
#! chage price to float to int
df['price'] =df['price'].astype('int')
```

```
#! changing the id and host id type of now
```

```
df['host_id']= df['host_id'].astype('int')  
df['id']= df['id'].astype('int')
```

```
#! date time  
df['last_review']=pd.to_datetime(df['last_review'])
```

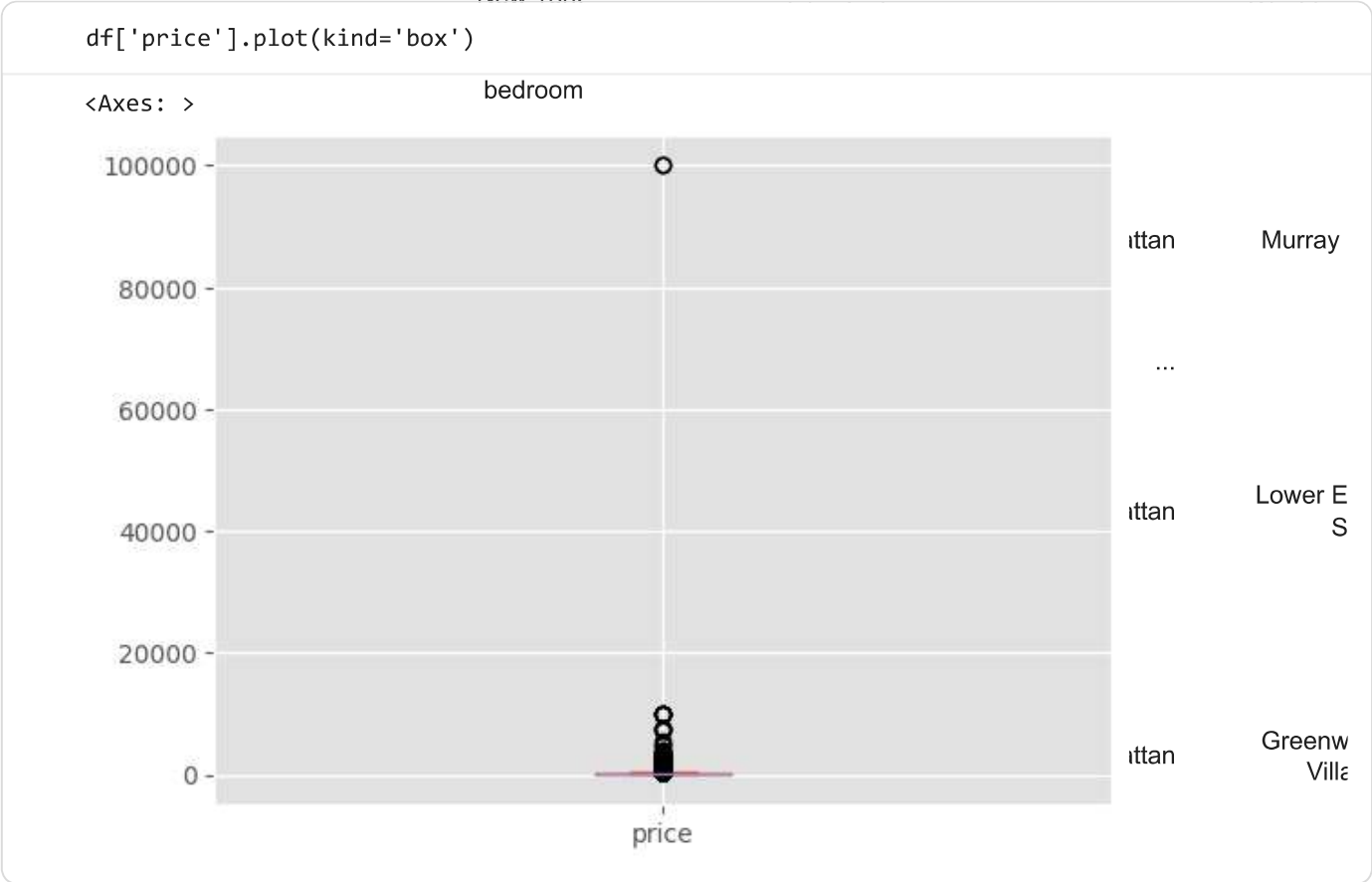
```
df
```


	id	name	host_id	host_name	neighbourhood_group	neighbourhood
0	1312228	Rental unit in Brooklyn · ★5.0 · 1 bedroom	7130382	Walter	Brooklyn	Clinton Hill
1	45277537	Rental unit in New York · ★4.67 · 2 bedrooms · ...	51501835	Jeniffer	Manhattan	Hell's Kitchen
2	9710000000000000000	Rental unit in New York · ★4.17 · 1 bedroom · ...	528871354	Joshua	Manhattan	Chelsea

```
df.shape
```

```
(20724, 7)
```

Univariate analysis



```
df['price'].skew()
```

```
20767.0
```

```
df[df['price']>50000]
```

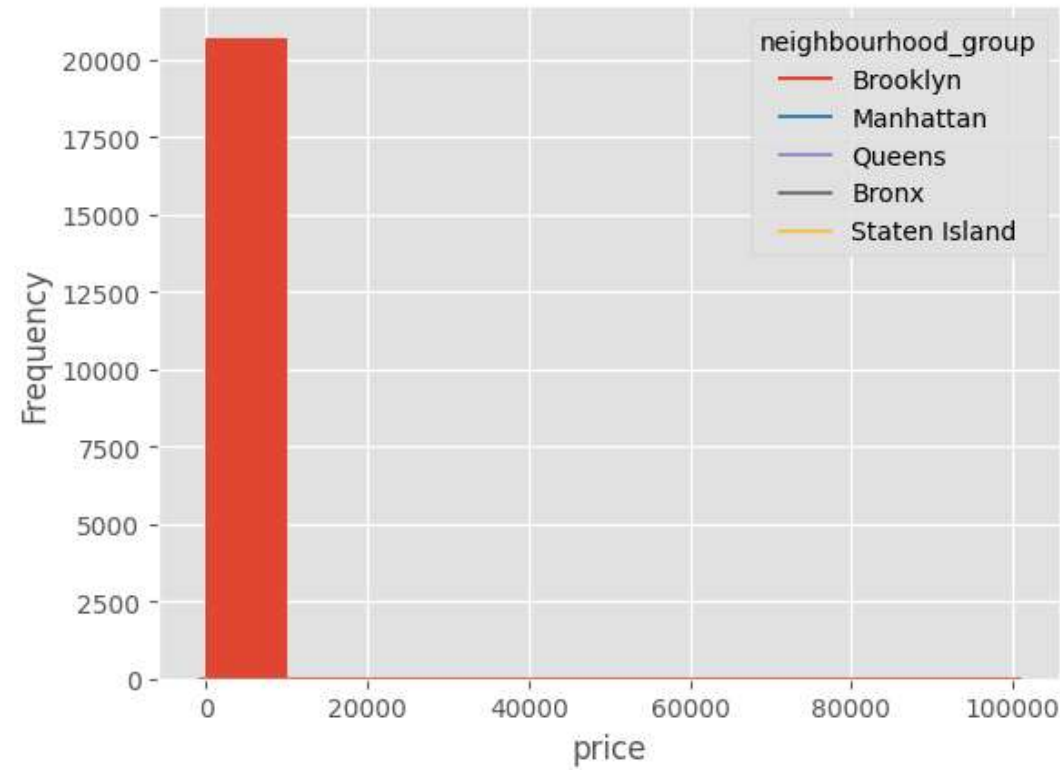
	id	name	unit in	host_id	host_name	neighbourhood_group	neighbourhood
20768	7830000000000000000	New York	unit in Manhattan	163083101	Marissa	Manhattan	Chinatown
3995	17160286	Brooklyn	Rental unit in Brooklyn · ★4.48 · 1 bedroom · ...	110361431	Bobbi	Brooklyn	Bedford-Stuyvesant
5497	6050000000000000000	Brooklyn	Rental unit in Brooklyn · ★4.33 · 1 bedroom · ...	110361431	Bobbi	Brooklyn	Bedford-Stuyvesant

20724 rows × 22 columns

2 rows × 22 columns

```
df['price'].plot(kind='hist')
sns.kdeplot(data=df,x='price',hue='neighbourhood_group')
```

<Axes: xlabel='price', ylabel='Frequency'>



```
df[df['price']>10000]
```

		id	name	host_id	host_name	neighbourhood_group	neighbourhood
3995		17160286	Rental unit in Brooklyn · ★4.48 · 1 bedroom · ...	110361431	Bobbi	Brooklyn	Bedford-Stuyvesant
5497	6050000000000000000		Rental unit in Brooklyn · ★4.33 · 1 bedroom · ...	110361431	Bobbi	Brooklyn	Bedford-Stuyvesant

2 rows × 22 columns

```
## price analysis
# d =df[df['price']>15000].dropna(inplace=True)
```

```
#! identyfing outliers in price
df=df[df['price']<1200]
```

```
df.shape

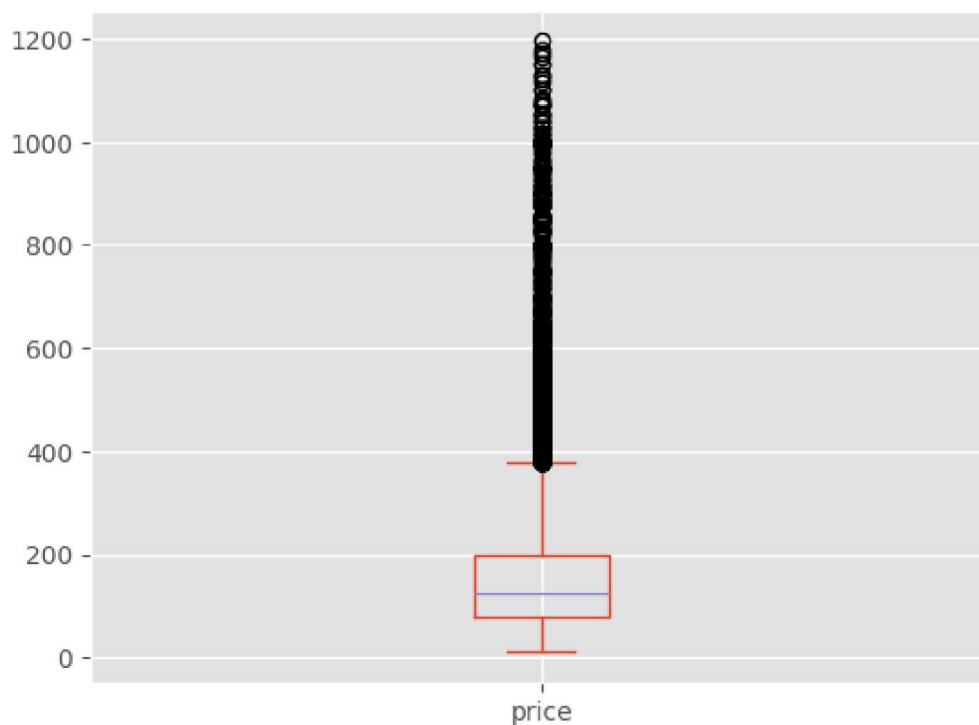
(20590, 22)
```

```
df['price'].sample(10)

4984    199
3362    129
4490    155
2060    323
5075    109
3216     90
7150     39
3929     67
8421    150
7973    225
Name: price, dtype: int64
```

```
df['price'].plot(kind='box')
```

<Axes: >



```
df['price'].skew()
```

```
np.float64(2.9071885092018275)
```

Removing outlier using upper limit one new thing

```
#!/ removing outliers
upper_limit = df['price'].quantile(0.98)

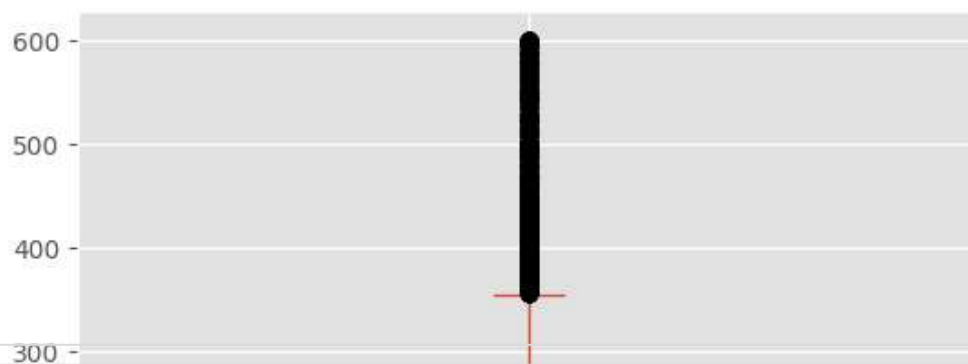
df= df[df['price'] <=upper_limit]
```

```
df['price'].skew()
```

```
np.float64(1.6385074790875476)
```

```
df['price'].plot(kind='box')
```

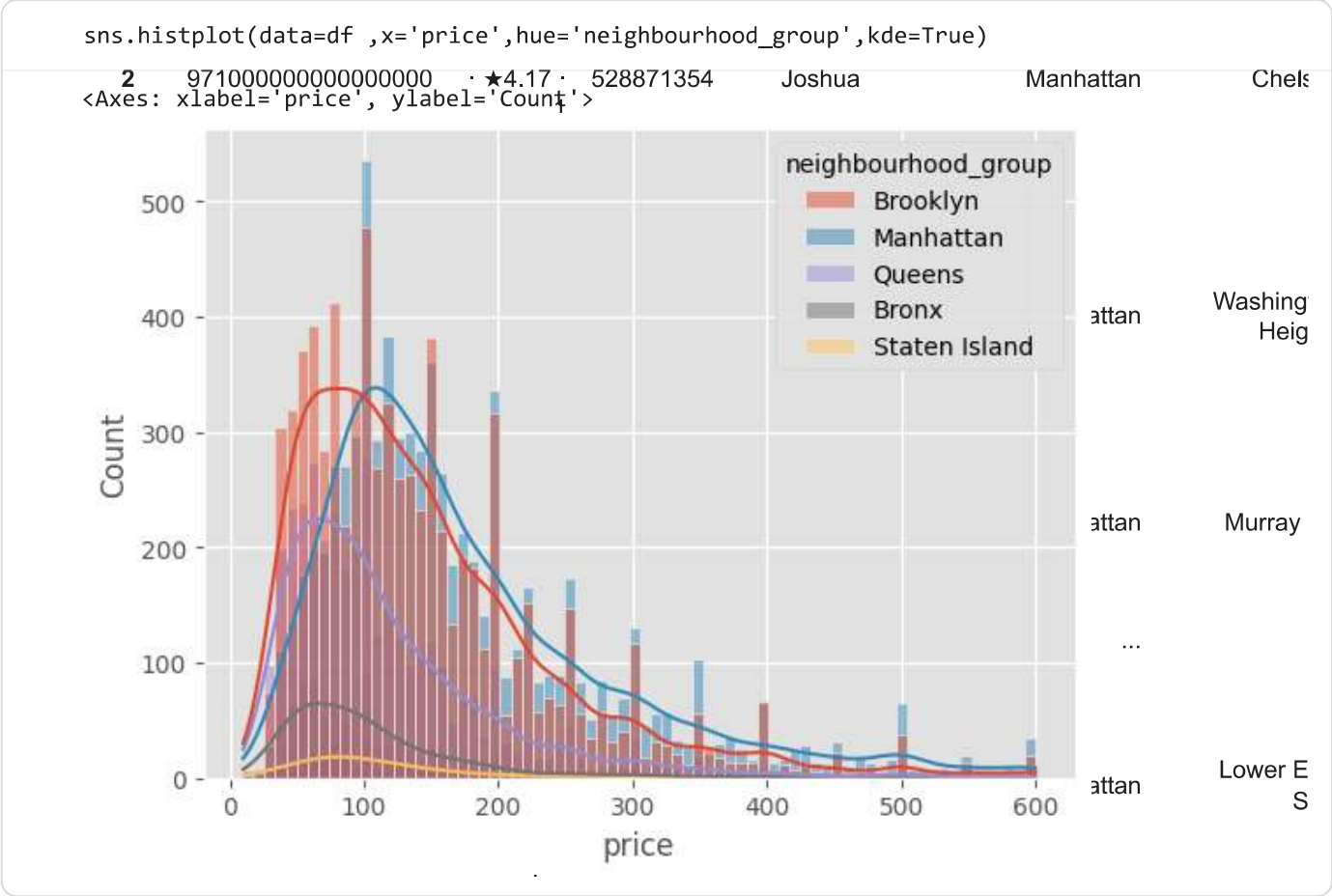
<Axes: >



df



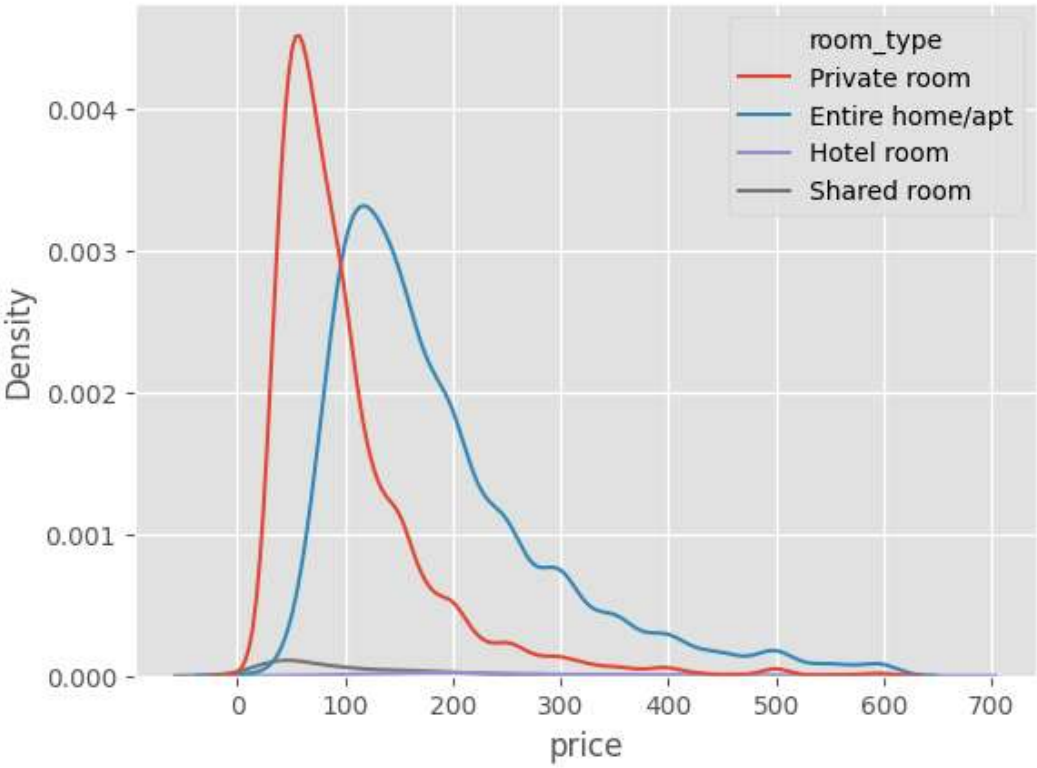
	id	name	host_id	host_name	neighbourhood_group	neighbourhc
0	1312228	Rental unit in Brooklyn · ★5.0 · 1 bedroom	7130382	Walter	Brooklyn	Clinton
1	45277537	Rental unit in New York · ★4.67 · 2 bedrooms · ...	51501835	Jeniffer	Manhattan	Hell's Kitch
		Rental				



```
sns.kdeplot(data=df,x='price',hue='room_type')
```

20766	2835711	New York · ★4.46 · 1 bedroom · ...	3237504	Aspen	Manhattan	Greenw Vill
20767	51825274	Rental unit in New York · ★4.93 · 1 bedroom · ...	304317395	Jeff	Manhattan	Hell's Kitch

<Axes: xlabel='price', ylabel='Density'>

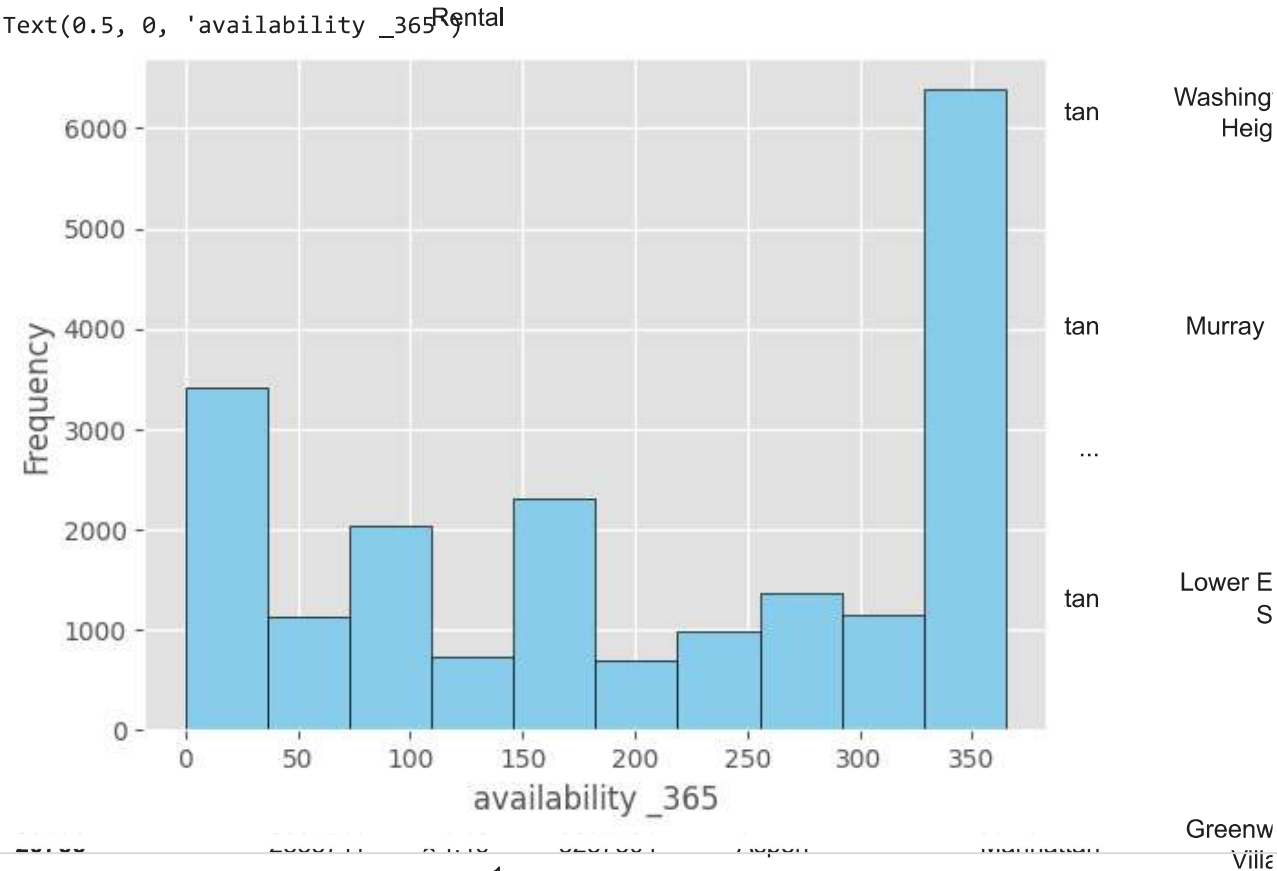


an Chinato
ns Rosed

df

	id	name	host_id	host_name	neighbourhood_group	neighbourho
0	1312228	Rental unit in Brooklyn · ★5.0 · 1 bedroom	7130382	Walter	Brooklyn	Clinton
1	45277537	Rental unit in New York · ★4.67 · 2 bedrooms · ...	51501835	Jeniffer	Manhattan	Hell's Kitch
		Rental				

```
#! minimum night
df['availability_365'].plot(kind='hist',edgecolor='black',color='skyblue')
plt.xlabel('availability_365')
```

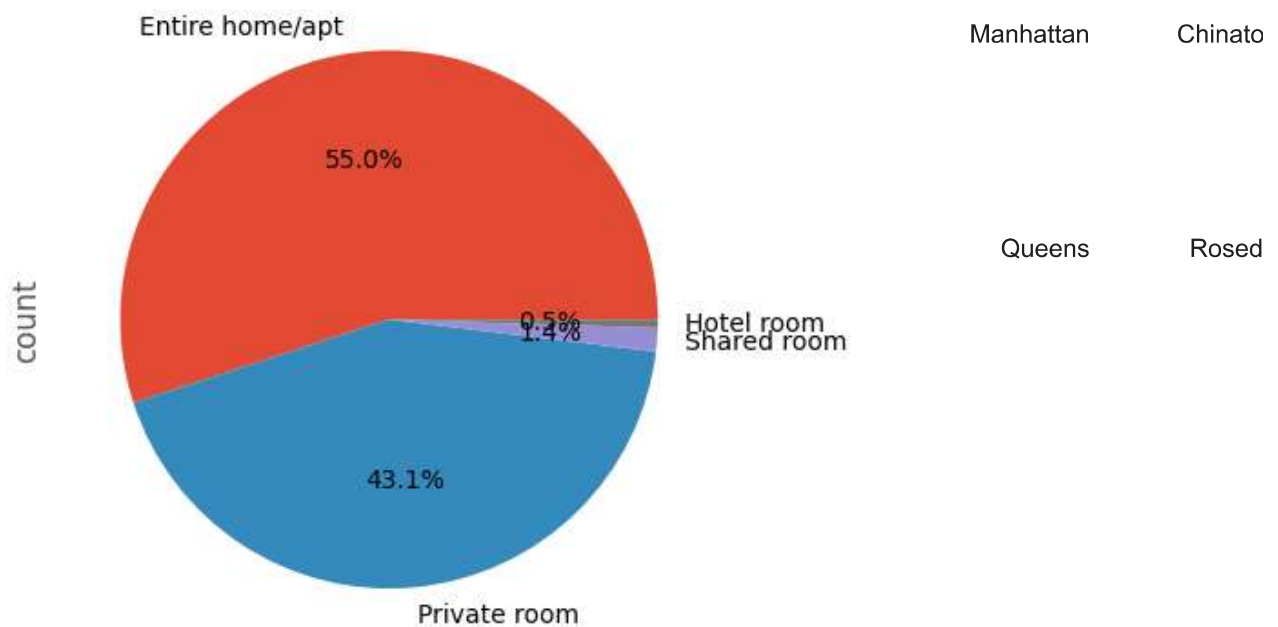


```
df['room_type'].value_counts().plot(kind='pie', autopct='%0.1f%%')
```

20767	51825274	Rental unit in New York · ★4.93 · 1 bedroom · ...	304317395	Jeff	Manhattan	Hell's Kitch
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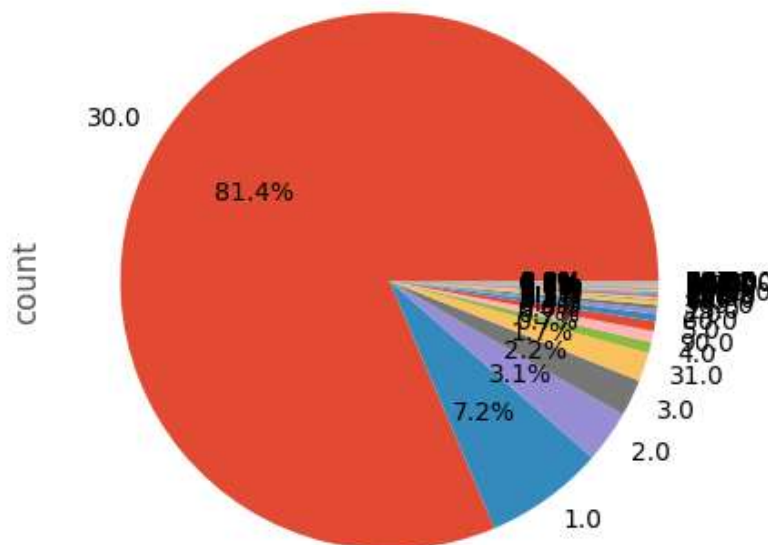
<Axes: ylabel='count'>

Rental



```
df['minimum_nights'].value_counts().plot(kind='pie', autopct="%0.1f%%")
```

<Axes: ylabel='count'>



df.columns

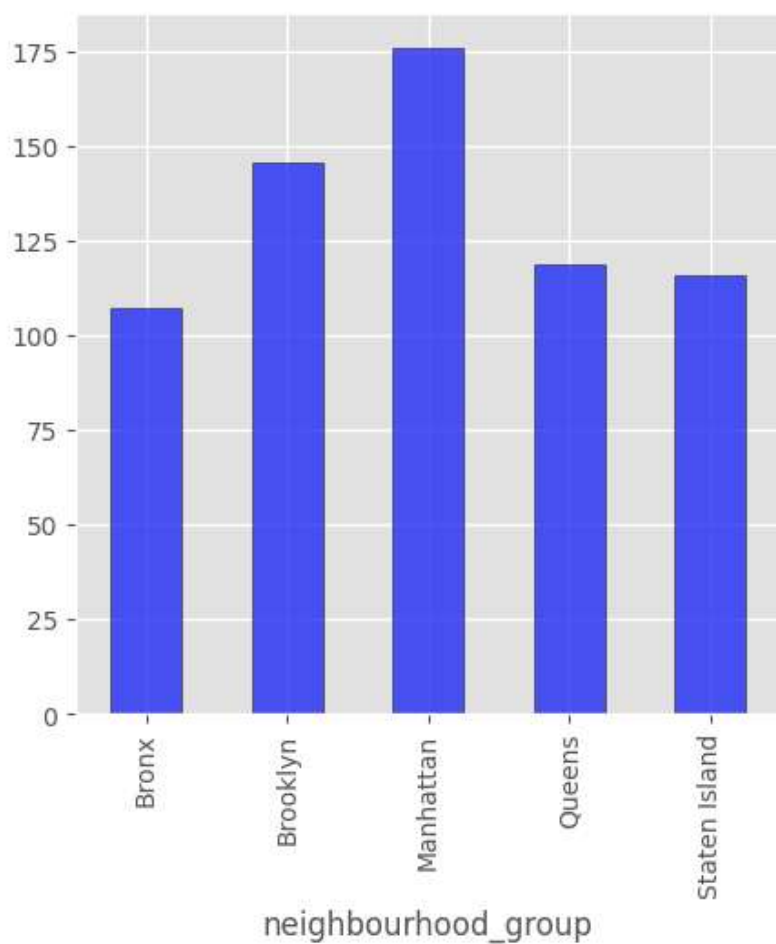
```
Index(['id', 'name', 'host_id', 'host_name', 'neighbourhood_group',
      'neighbourhood', 'latitude', 'longitude', 'room_type', 'price',
      'minimum_nights', 'number_of_reviews', 'last_review',
      'reviews_per_month', 'calculated_host_listings_count',
      'availability_365', 'number_of_reviews_ltm', 'license', 'rating',
      'bedrooms', 'beds', 'baths'],
      dtype='object')
```

✓ bivariate analysis

```
#! neighbour hood and price group by to annalyized
```

```
df.groupby('neighbourhood_group')['price'].mean().plot(kind='bar',color="#0911F9",edgecolor=
```

```
<Axes: xlabel='neighbourhood_group'>
```



✓ per bed Feature Engineerring

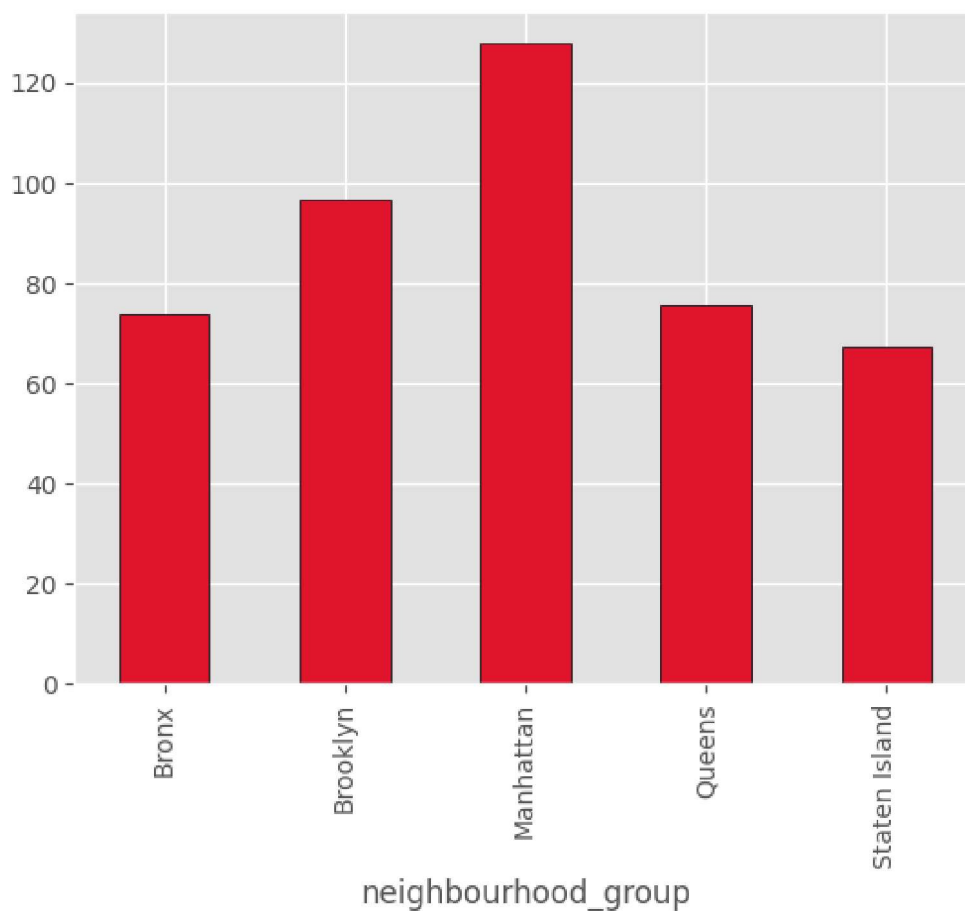
```
df['per_bed'] = df['price']/df['beds']
```

```
df.groupby('neighbourhood_group')['per_bed'].mean().reset_index()
```

	neighbourhood_group	per_bed
0	Bronx	73.736568
1	Brooklyn	96.766033
2	Manhattan	127.730956
3	Queens	75.712101
4	Staten Island	67.271991

```
df.groupby('neighbourhood_group')['per_bed'].mean().plot(kind='bar',color="#E11530",edgecolor
```

```
<Axes: xlabel='neighbourhood_group'>
```

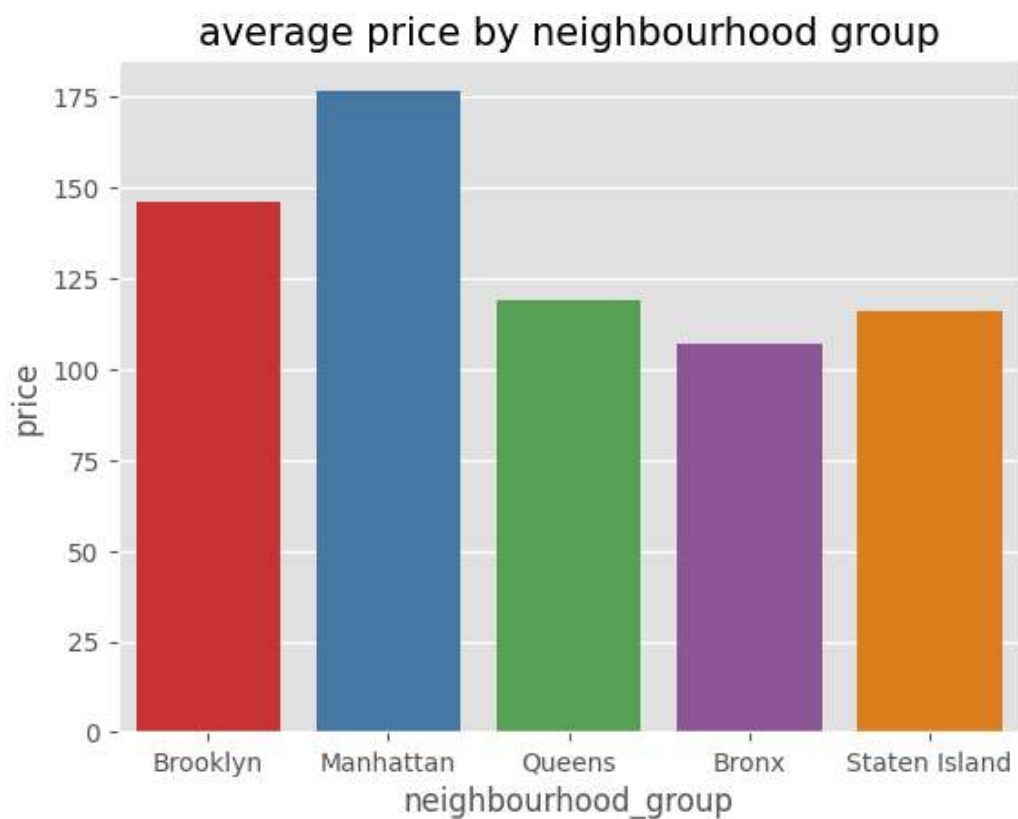


✓ bivariate analysis

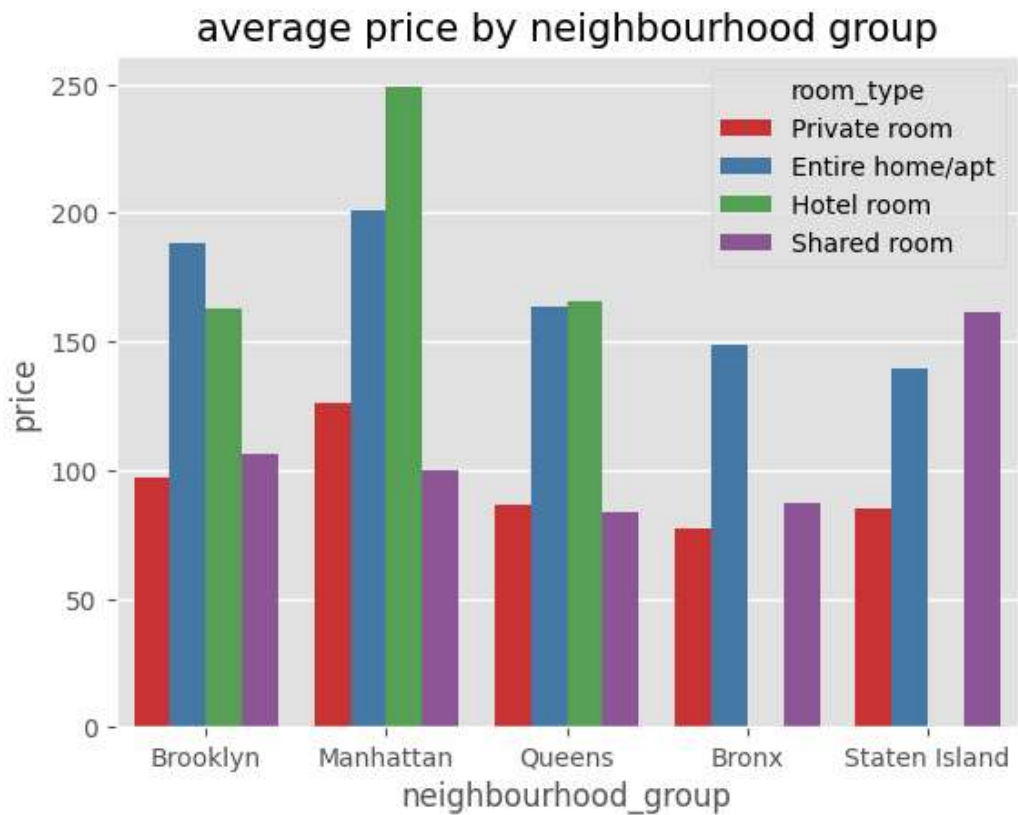
```
df.columns
```

```
Index(['id', 'name', 'host_id', 'host_name', 'neighbourhood_group',
      'neighbourhood', 'latitude', 'longitude', 'room_type', 'price',
      'minimum_nights', 'number_of_reviews', 'last_review',
      'reviews_per_month', 'calculated_host_listings_count',
      'availability_365', 'number_of_reviews_ltm', 'license', 'rating',
      'bedrooms', 'beds', 'baths', 'per_bed'],
      dtype='object')
```

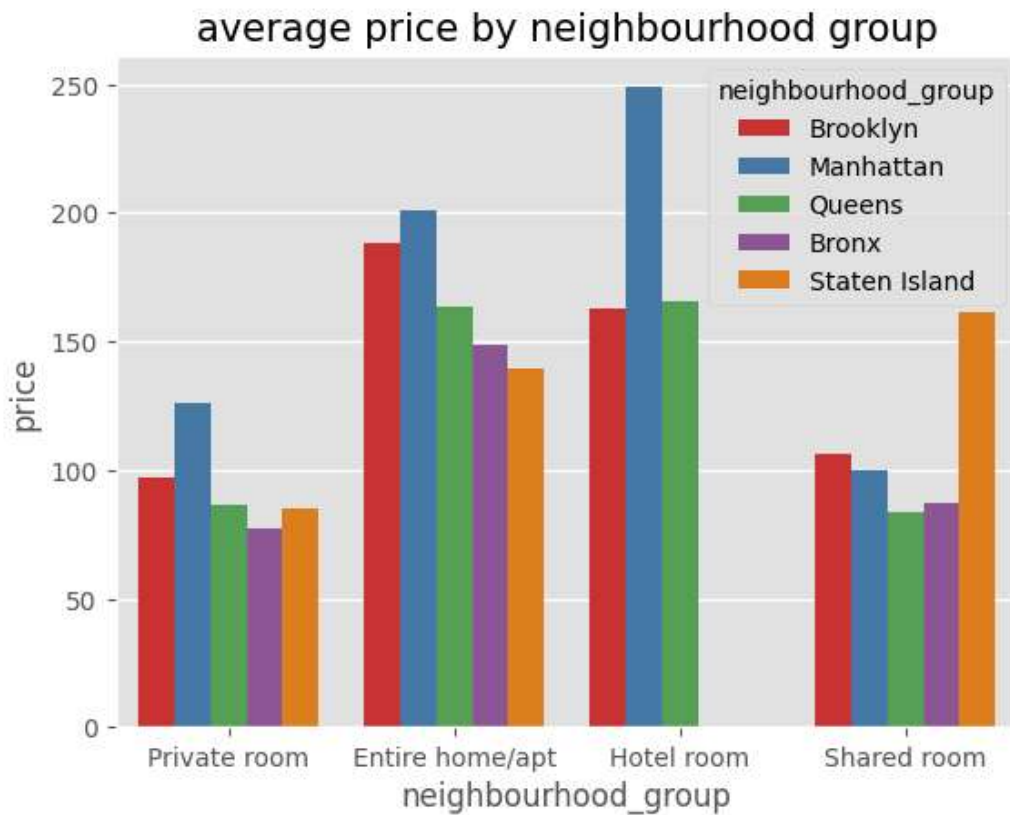
```
sns.barplot(data=df,x='neighbourhood_group',y='price',errorbar=None,estimator=np.mean,hue='i
plt.title("average price by neighbourhood group",fontsize='15',weight='light')
plt.xlabel("neighbourhood_group")
plt.ylabel('price')
plt.show()
```



```
sns.barplot(data=df,x='neighbourhood_group',y='price',errorbar=None,estimator=np.mean,hue='i
plt.title("average price by neighbourhood group",fontsize='15',weight='light')
plt.xlabel("neighbourhood_group")
plt.ylabel('price')
plt.show()
```

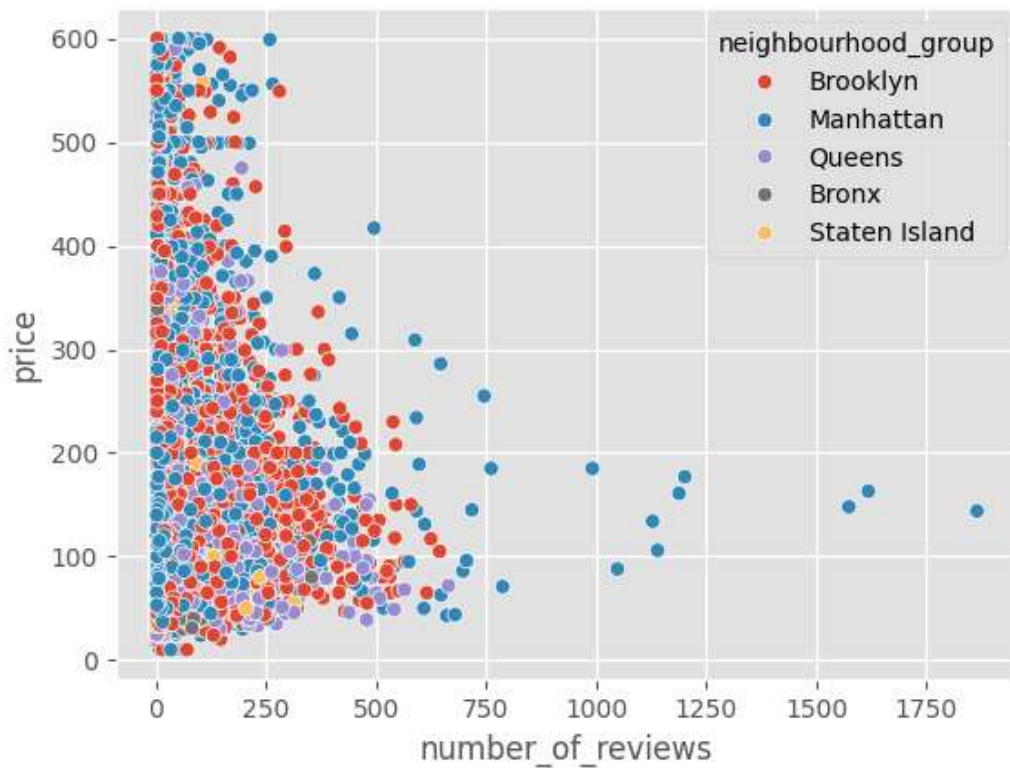


```
sns.barplot(data=df,x='room_type',y='price',errorbar=None,estimator=np.mean,hue='neighbourh  
plt.title("average price by neighbourhood group",fontsize='15',weight='light')  
plt.xlabel("neighbourhood_group")  
plt.ylabel('price')  
plt.show()
```



```
sns.scatterplot(data=df,y='price',x='number_of_reviews',hue='neighbourhood_group')
```

```
<Axes: xlabel='number_of_reviews', ylabel='price'>
```

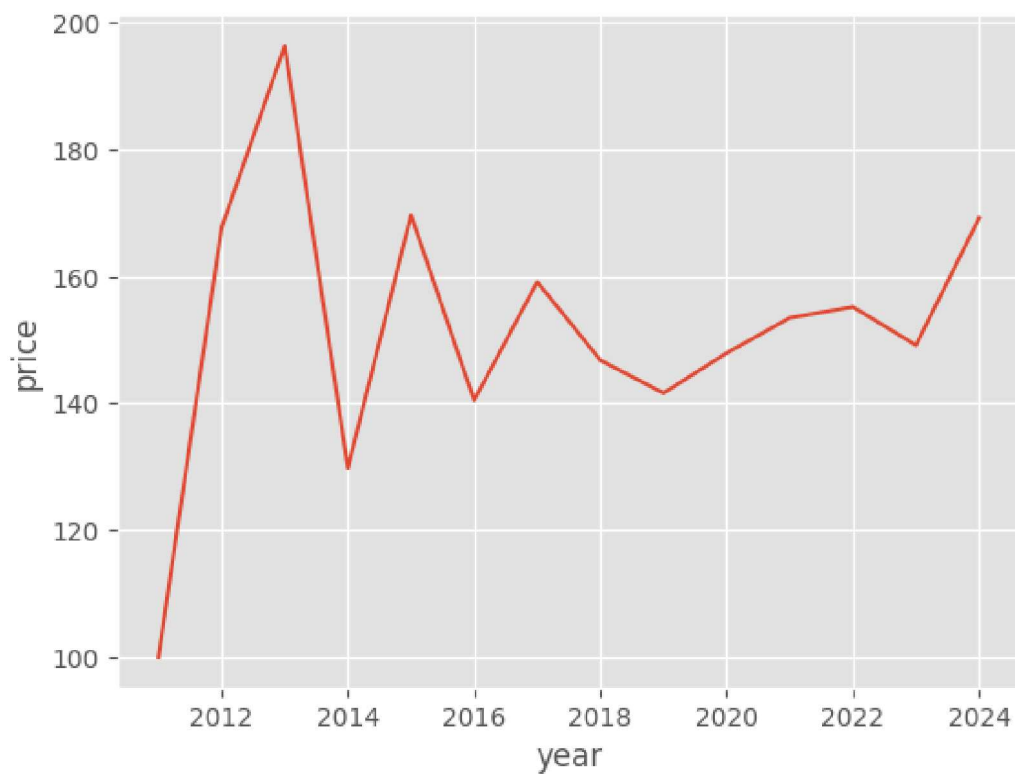


```
df['year'] = df['last_review'].dt.year
```

```
#! line plot trend over year like avg price per year
```

```
avg_price= df.groupby('year')['price'].mean().reset_index()  
sns.lineplot(data=avg_price,x='year',y='price',palette='Set3')
```

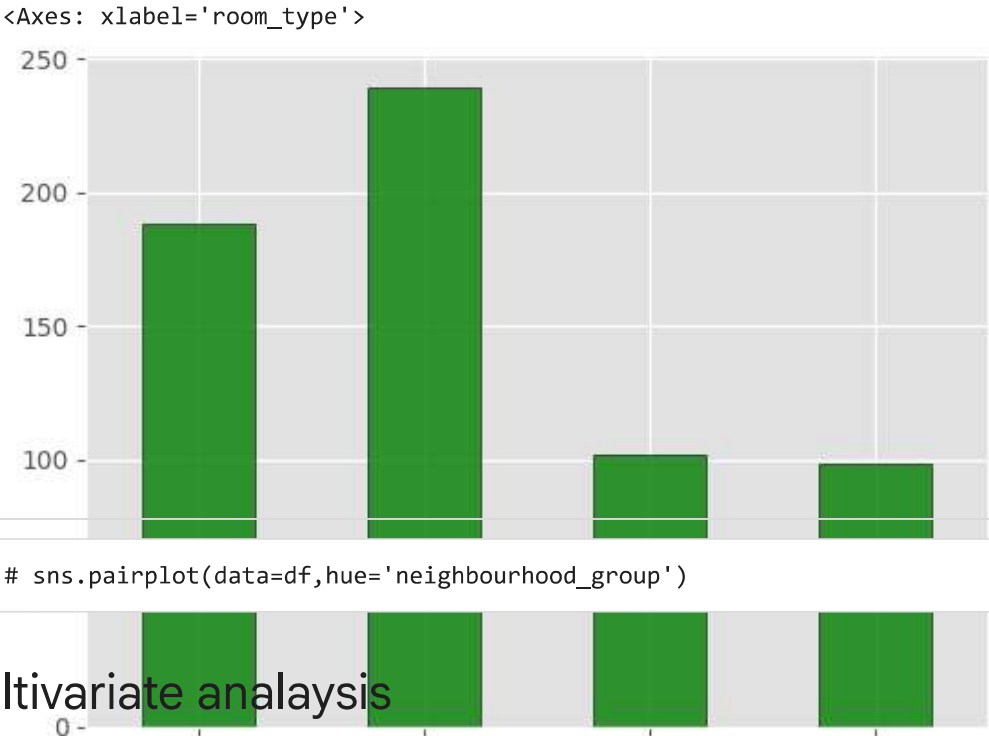
<Axes: xlabel='year', ylabel='price'>



```
df.groupby('room_type')['price'].mean()
```

```
room_type  
Entire home/apt    188.521055  
Hotel room         239.564356  
Private room       101.628663  
Shared room        98.582456  
Name: price, dtype: float64
```

```
df.groupby('room_type')['price'].mean().plot(kind='bar',color='green',alpha=0.8,edgecolor='b1
```



▼ Multivariate analysis



	id	name	host_id	host_name	neighbourhood_group	neighbourhood
0	1312228	Rental unit in Brooklyn · ★5.0 · 1 bedroom	7130382	Walter	Brooklyn	Clinton
1	45277537	Rental unit in New York · ★4.67 · 2 bedrooms · ...	51501835	Jeniffer	Manhattan	Hell's Kitc
		Rental				

```
sns.pairplot(data=df,vars=['price','minimum_nights','number_of_reviews','availability_365']).plt.figure(figsize=(10,10))
```



<Figure size 1000x1000 with 0 Axes>

20767	51825274	Rental unit in New York · ★4.93 · 1 bedroom	304317395	Jeff	Manhattan	Hell's Kitc
-------	----------	---	-----------	------	-----------	-------------

```
df.columns
```

```
Index(['id', 'name', 'host_id', 'unit in New York', 'host_name', 'neighbourhood_group',
      'lat', 'lon', 'room_type', 'price', 'Manhattan', 'Chinato',
      'minimum_nights', 'number_of_reviews', 'last_review',
      'reviews_per_month', 'calculated_host_listings_count',
      'availability_365', 'number_of_reviews_ltm', 'license', 'rating',
      'bedrooms', 'beds', 'baths', 'rental', 'per_bed', 'year'],
      dtype='object')
```

```

#* geographical distrubutation of airbnb listing
plt.figure(figsize=(10,10))
sns.scatterplot(data=df,x='longitude',y='latitude',hue='room_type')

```

```
<Axes: xlabel='longitude', ylabel='latitude'>
```

